

ECE 100 – Linear Electronic Systems

Lab 2 Report – Active Circuit Design April 24, 2026

Diego Ceja

PID: A17329002

Professor: Saharnaz Baghdadchi

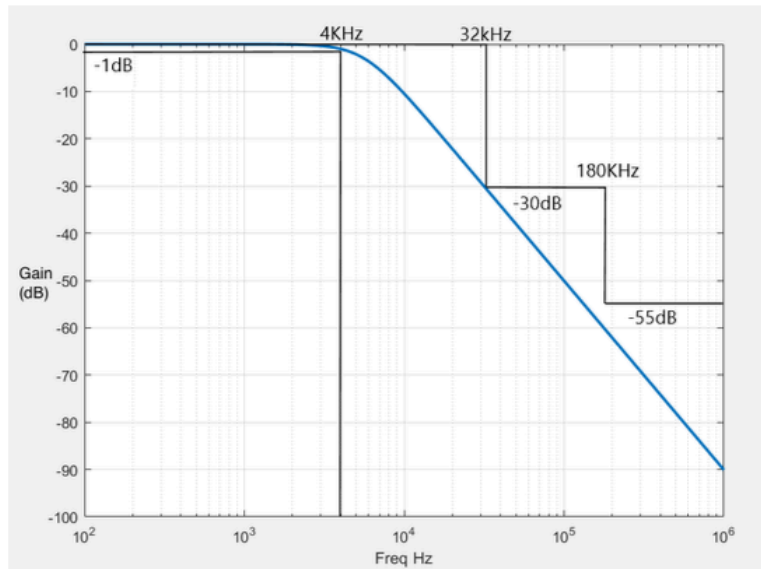
Table of Contents

Abstract.....	3
Experimental Procedures and Results.....	4
1: Approximation.....	4
2: Realization/Simulation.....	7
3: Experiment.....	9
Conclusion.....	11
MATLAB.....	12

Abstract

The purpose of this lab is to analyze active circuits, which are circuits with energy supplying components in them. Through this, we will simulate a low-pass filter, with our goal being to separate a digital DSL signal from the voice signal on a telephone line.

Experimental Procedures and Results



Problem 1: Approximation

To begin our lab, we were given a simple circuit as we want to reduce complexity and cost. We start off with a Butterworth filter for convenience, with an equation $|H(f)|^2 = 1/(1+(f/f_0)^4)$. We need to see if this filter meets the specifications.

a)

1a) 2nd Order Butterworth Filter

$$|H(f)|^2 = \frac{1}{1 + (\frac{f}{f_0})^4}$$

$$\Rightarrow .7943 = \frac{1}{1 + (\frac{f}{f_0})^4}$$

$$w/ f = 4 \text{ kHz} \quad .7943(1 + (\frac{4 \text{ kHz}}{f_0})^4) = 1$$

$$\sqrt[4]{(\frac{4 \text{ kHz}}{f_0})^4} = \sqrt[4]{.25897}$$

$$(\frac{4 \text{ kHz}}{f_0}) = .713366$$

$$10 \log |H(f)|^2 = -1 \text{ dB}$$

$$\log |H(f)|^2 = \frac{-1}{10}$$

$$|H(f)|^2 = 10^{-1/10} = .7943$$

For 32 kHz, 180 kHz

$$f_0 = 5607.22 \text{ Hz} \Rightarrow 5.61 \text{ kHz}$$

$$|H(32 \text{ kHz})|^2 = \frac{1}{1 + (\frac{32 \text{ kHz}}{5.61 \text{ kHz}})^4} = 9.437 \times 10^{-4}$$

$$10 \log |H(f)|^2 \Rightarrow 10 \log (9.437 \times 10^{-4}) = -30.25 \text{ dB}$$

$$-30.25 < -30 \quad \checkmark$$

$$|H(180 \text{ kHz})|^2 = \frac{1}{1 + (\frac{180 \text{ kHz}}{5.61 \text{ kHz}})^4} = 9.435 \times 10^{-7}$$

$$10 \log (9.435 \times 10^{-7}) = -60.25 \text{ dB}$$

$$-60.25 < -55 \quad \checkmark$$

Filter meets spec

b)

The filter slightly meets the specifications, but this can be improved. First, we will ripple the filter from -1dB to 0dB, then down to -1dB at 4 kHz. We now need to find f_{peak} , and find the dampening ratio ζ

$$1b) \quad H(j\omega) = \frac{(.7943)^{1/2}}{1 + 2\zeta(\frac{j\omega}{f_0}) + (\frac{j\omega}{f_0})^2} = \frac{(.7943)^{1/2}}{1 - (\frac{f}{f_0})^2 + 2\zeta(\frac{f}{f_0})} \Rightarrow |H(f)|^2 = \frac{.7943}{(1 - (\frac{f}{f_0})^2)^2 + (2\zeta(\frac{f}{f_0}))^2} \quad \text{Let } \frac{f}{f_0} = x$$

$$1 - 2x^2 + x^4 + 4\zeta^2 x^2 \Rightarrow \frac{d}{dx} [1 - 2x^2(1 - 2\zeta^2) + x^4] = -4x(1 - 2\zeta^2) + 4x^3 = 0$$

$$\Rightarrow \sqrt{1 - 2\zeta^2} = \sqrt{x^2} \Rightarrow \frac{f}{f_0} \Rightarrow f_{\text{peak}} = f_0 \sqrt{1 - 2\zeta^2} \quad \left(\frac{f}{f_0}\right)^2 = 1 - 2\zeta^2$$

$$|H(0)|^2 = \frac{.7943}{1} = .7943 \quad |H(f_{\text{peak}})|^2 = \frac{.7943}{(1 - (1 - 2\zeta^2))^2 + 4\zeta^2(1 - 2\zeta^2)}$$

$$= \frac{.7943}{4\zeta^4 + 4\zeta^2 - 8\zeta^4} = \frac{.7943}{4\zeta^2(1 - 2\zeta^2)}$$

so for the ratio $\frac{|H(f_{\text{peak}})|^2}{|H(0)|^2} = \frac{.7943}{4\zeta^2(1 - 2\zeta^2)}$

$$\frac{|H(f_{\text{peak}})|^2}{|H(0)|^2} = \frac{1}{4\zeta^2(1 - 2\zeta^2)}$$

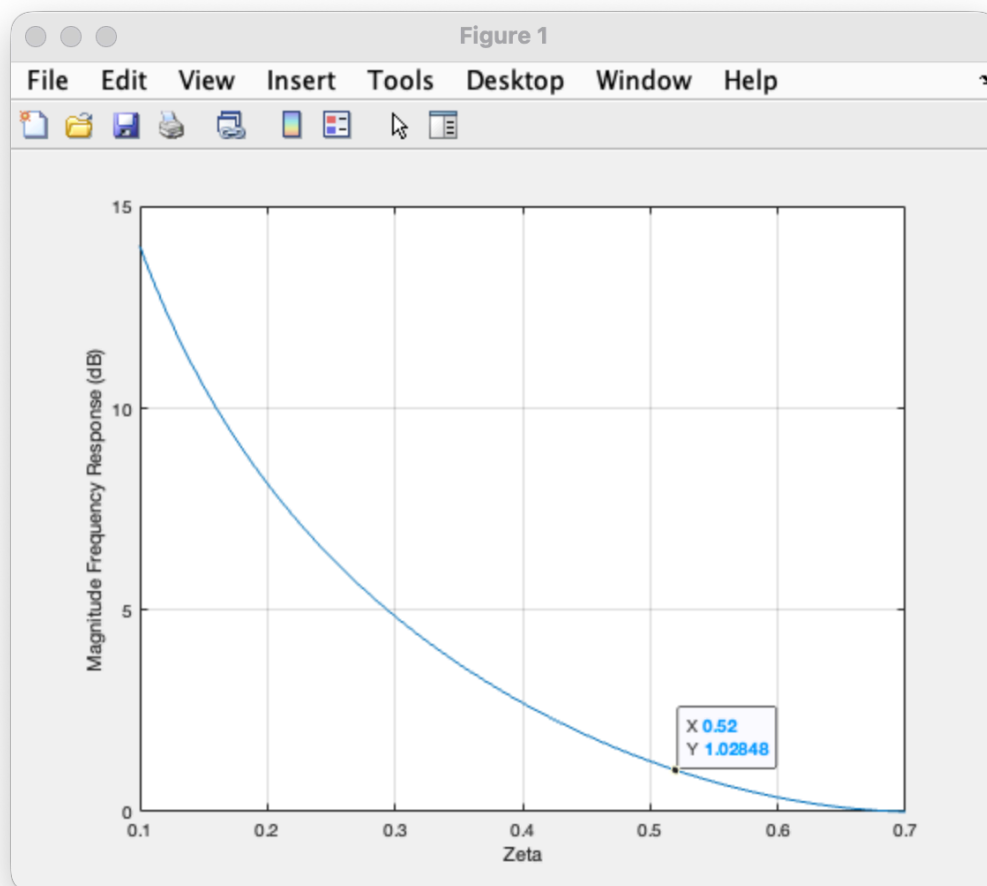


Figure 1: $|H(f_{\text{peak}})|^2 / |H(0)|^2$ vs ζ plot

c)

We have now obtained a value of zeta with a 1dB peak. Using MATLAB, we can approximate a better f_0

$$\begin{aligned} \text{1c)} \quad f_0 &= 5000 \times \frac{4000}{X} \quad \text{My } X \text{ was } \approx 4762.38 \\ \text{so } f_0 &= 5000 \times \frac{4000}{4762.38} \approx 4200 \text{ Hz} \end{aligned}$$

Through guessing a couple of f_0 's, we found its respective values where it has -1dB and -30dB.

d)

$$\begin{aligned} \text{1d)} \quad &\text{After testing, found a good } f_0 \text{ to be } = 4891 \text{ Hz}, \quad X_1 = 1.1647 \quad X_2 = 1.2228 \\ &\frac{|X_1 - X_2|}{X_1} = \frac{|1.1647 - 1.2228|}{1.1647} \times 100 = 4.99\% < 10\% \quad \text{Meets spec} \end{aligned}$$

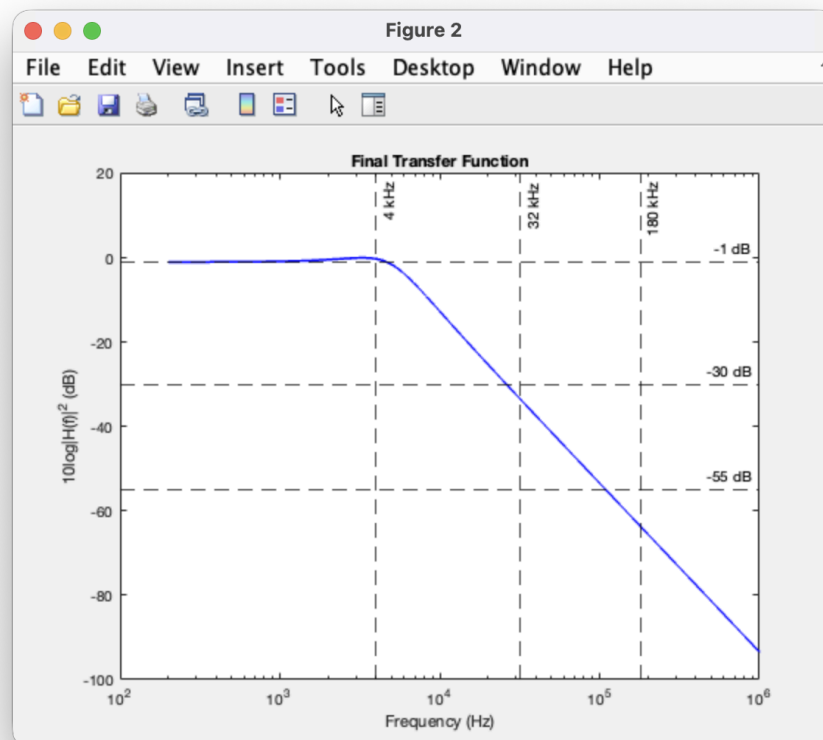


Figure 2: $10\log|H(f)|^2 \text{ dB}$ vs Frequency plot

Problem 2: Realization/Simulation

Let us realize this circuit and create a Sallen-Key form low pass filter. We use the Sallen-Key filter functions, with $R_1=R_2=100k$. We also keep in mind the -1dB Thevenin resistance. We also consider the fact that a non-ideal circuit has about 50 ohms, so we calculated this accordingly.

2a) From part 1, $\zeta = .5227$ $\omega_0 = 4.891 kHz$ $H(s)_{Sallen-Key} = \frac{1}{1 + sC_1(R_1 + R_2) + s^2 C_1 C_2 R_1 R_2} = \frac{1}{1 + s(\underbrace{C_1(R_1 + R_2)}_{\downarrow}) + s^2 (C_1 C_2 R_1 R_2)}$

2nd order form $H(s) = \frac{1}{1 + 2\zeta \frac{s}{\omega_0} + (\frac{s}{\omega_0})^2} = \frac{1}{1 + s(\frac{2\zeta}{\omega_0}) + s^2(\frac{1}{\omega_0^2})} \Rightarrow \frac{2\zeta}{\omega_0} = C_1(R_1 + R_2) \quad R_1 = R_2 = R \text{ so}$

$\Rightarrow \frac{2\zeta}{\omega_0} = C_1 \cdot 2R \rightarrow C_1 = \frac{\zeta}{\omega_0 R}$

$\Rightarrow \frac{1}{\omega_0^2} = C_1 C_2 R_1 R_2 \Rightarrow (\frac{\zeta}{\omega_0 R}) C_2 R^2 = \frac{\zeta C_2 R}{\omega_0^2} = \frac{1}{\omega_0^2}$

$\Rightarrow C_2 = \frac{1}{\omega_0^2 R} \quad C_1 = \frac{\zeta}{\omega_0 R}$

$\zeta = .5227 \quad R = 100k \quad \omega_0 = 2\pi f_0 = 30.731k \quad C_2 = 6.23 \times 10^{-10} F \quad C_1 = 1.70 \times 10^{-10} F$

$= .623 nF \quad = .170 nF$

$10 \log |H(s)|^2 = -1 \Rightarrow \log |H(s)|^2 = \frac{-1}{10}$

$\Rightarrow |H(s)|^2 = 10^{-1/10} = \sqrt{.7943}$

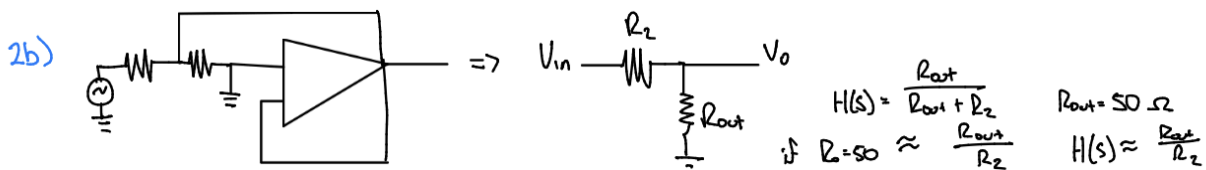
$|H(s)| = .891 \quad \text{so} \quad H(s) = \frac{R_b}{R_b + R_a} \Rightarrow .891 = \frac{R_b}{R_b + R_a}$

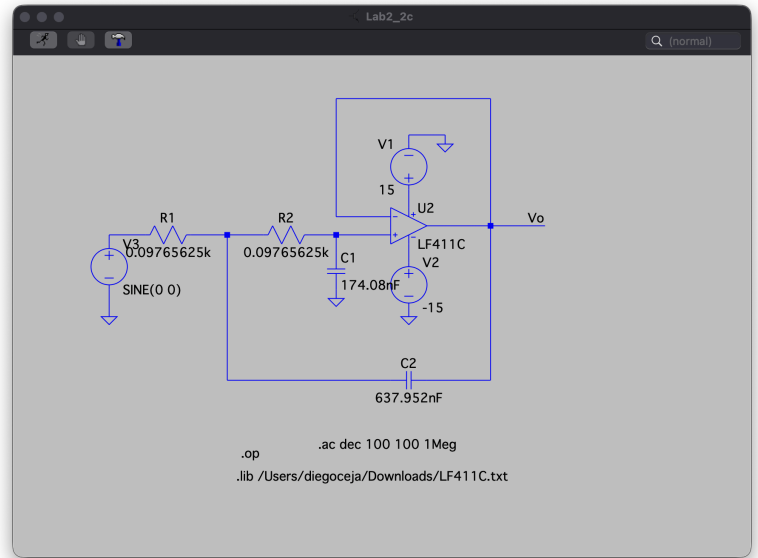
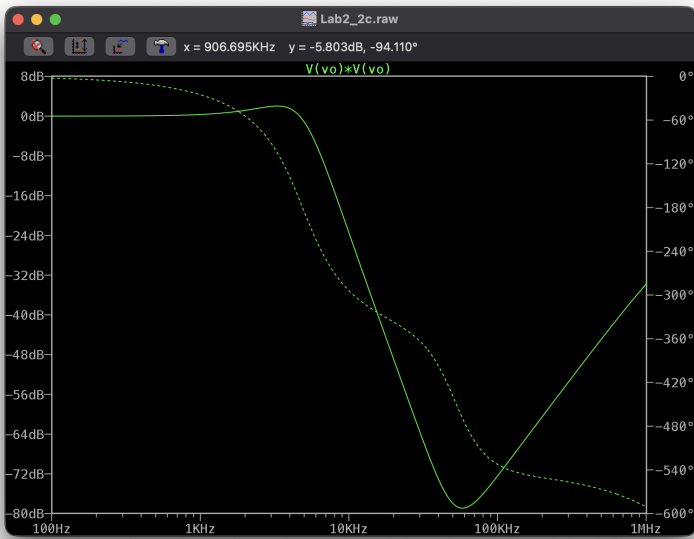
$\Rightarrow .891(R_b + R_a) = R_b \quad R_a = .122 R_b$

$R_a || R_b = \frac{R_a R_b}{R_a + R_b} = 100k \quad \hookrightarrow \frac{(.122 R_b) R_b}{.122 R_b + R_b} = \frac{.122 R_b^2}{1.122 R_b} = .1087 R_b = 100k$

$R_b = 919.67k \Omega$

So $R_a = .122(919.67k) = 122.2k \Omega$





After some testing and simulating, i found out that the lowest resistance that was still working was $0.09765625\text{k}\Omega$

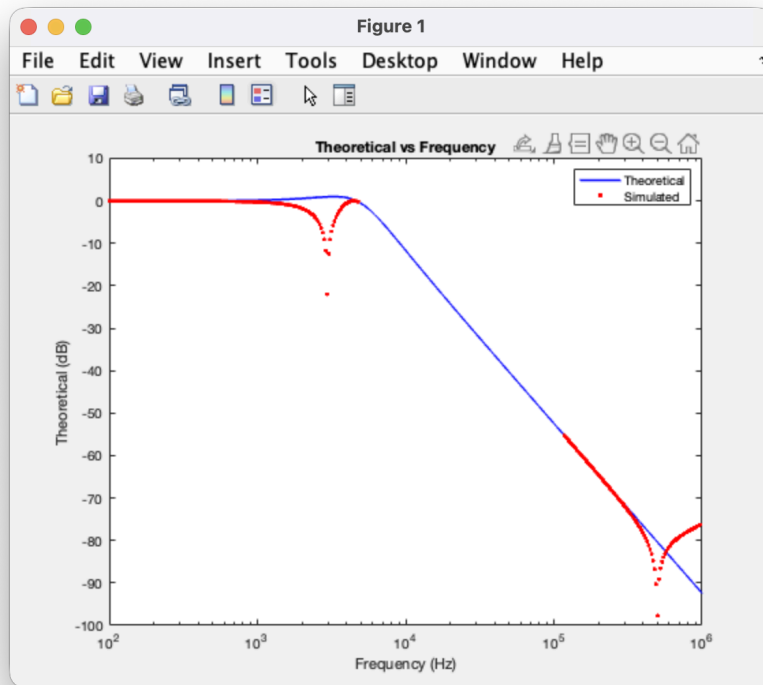


Figure 3: Theoretical vs Frequency plot

Problem 3: Experiment

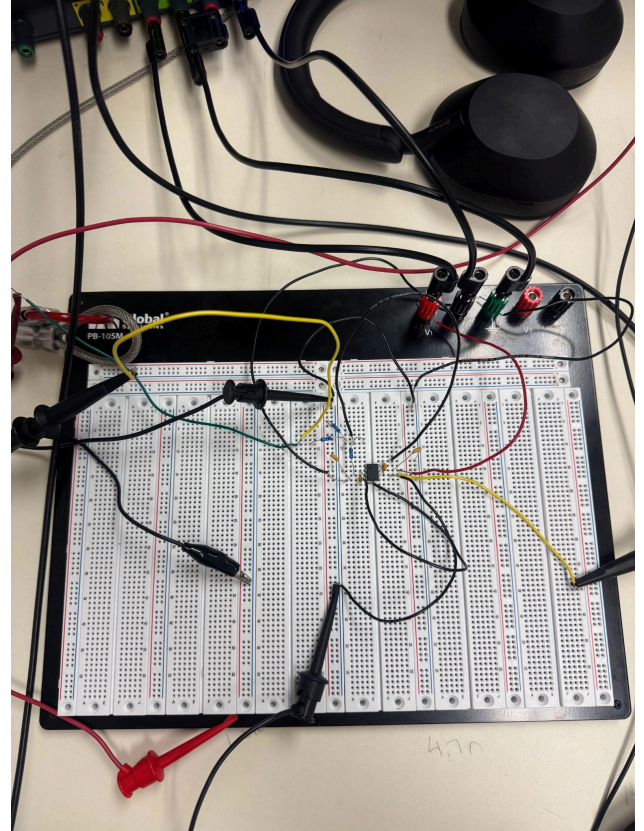
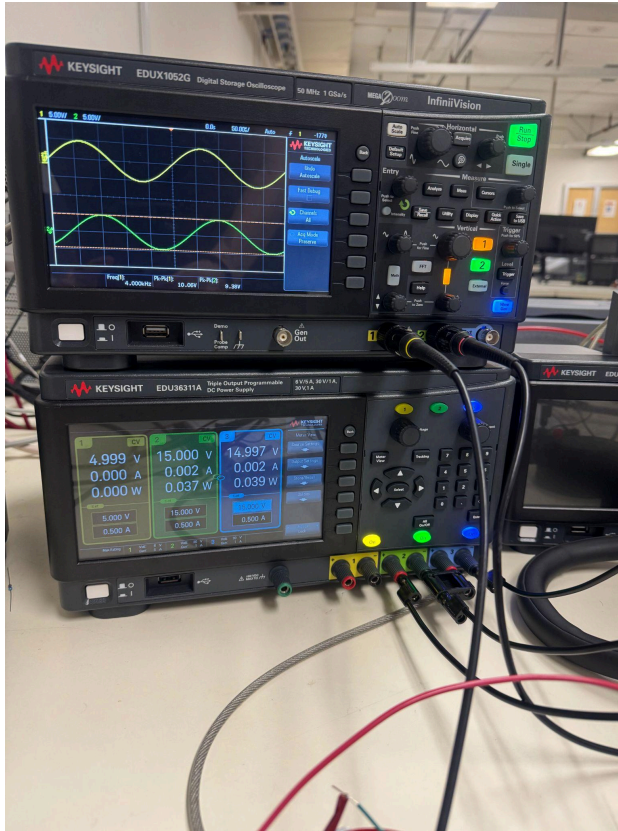


Figure 4 and 5: Signal Waveform at 4 kHz and Circuit Board

Frequency (Hz)	Input Vpp	Output Vpp	dB
10 Hz	9.5 V	9.09 V	-0.383
4 kHz	10 V	9.4 V	-0.537
$f_{\text{peak}} = 4.891 \text{ kHz}$	10 V	8.89 V	-1.022
32 kHz	10 V	24 mV	-30
180 kHz	10 V	25 mV	-66.94
1 MHz	10 V	135 mV	-38.03

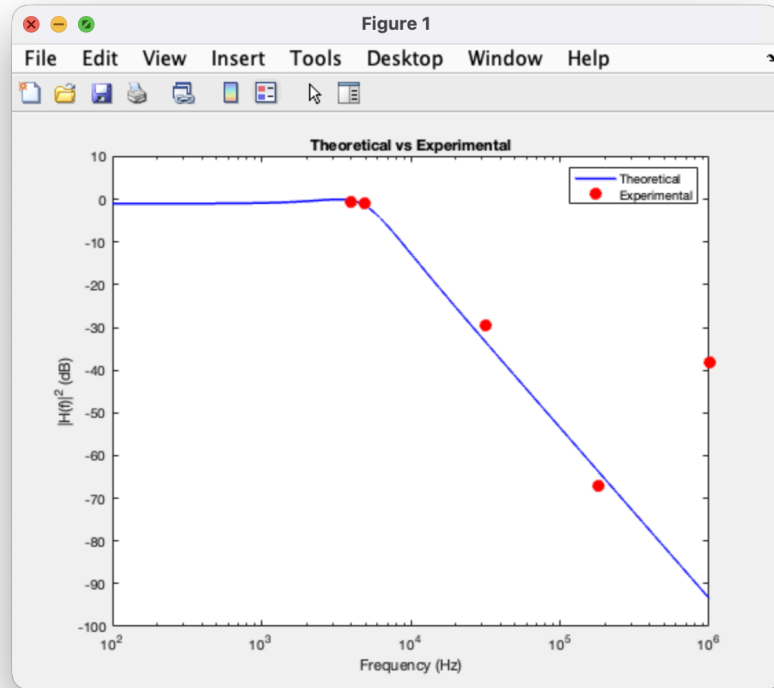
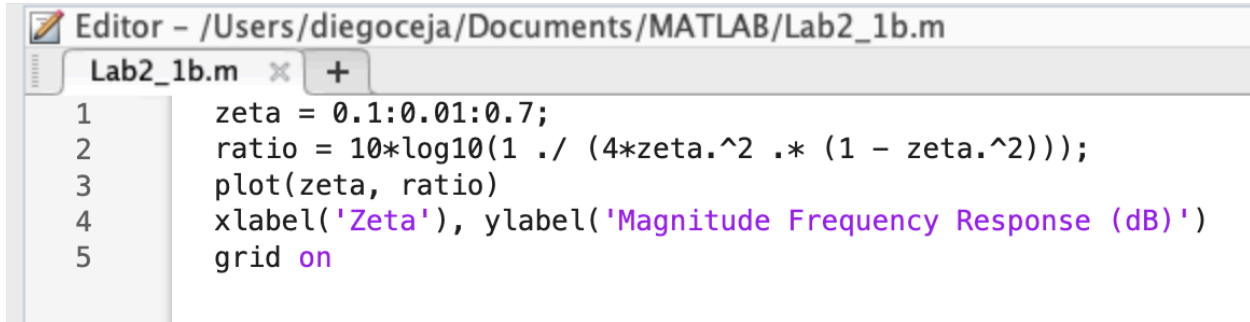


Figure 6: Theoretical Waveform vs My Experimental Points

Conclusion

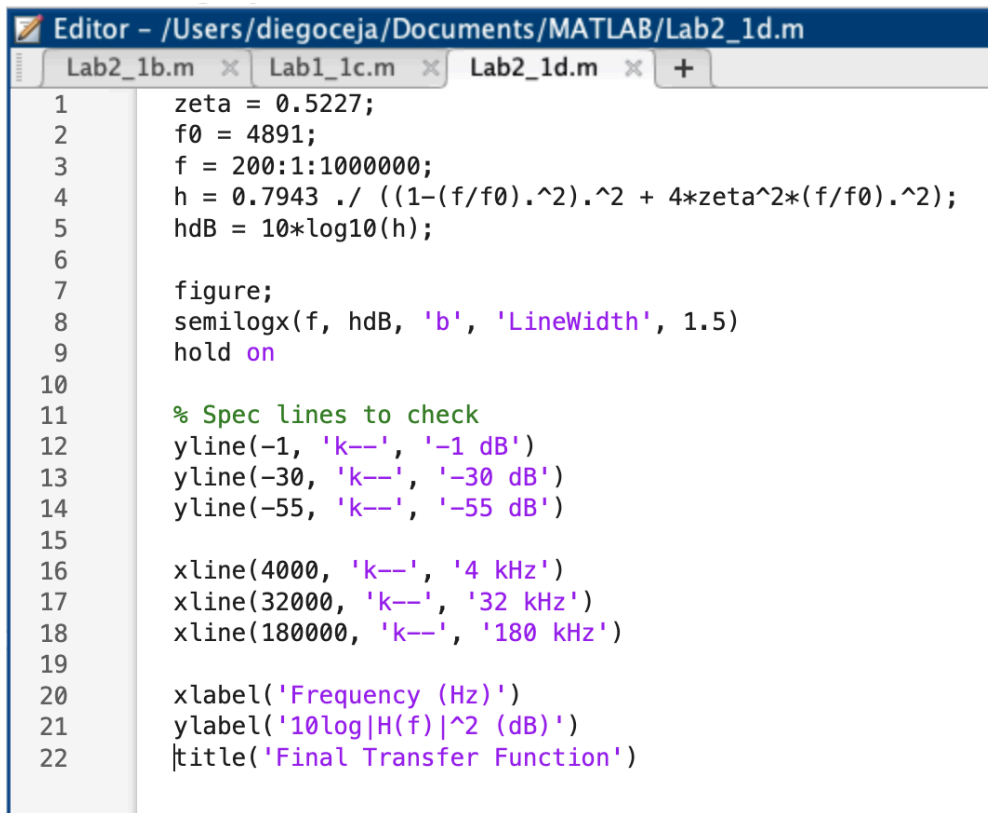
In conclusion, the Sallen-Key low-pass filter is useful for separating a source signal from unwanted noise, allowing us as users to clearly hear those who need to be heard. Though we had some minor understandings of op-amps in ECE 65, we saw a greater application of one in this lab.

MATLAB Code



```
Editor - /Users/diegoceja/Documents/MATLAB/Lab2_1b.m
Lab2_1b.m x +
1 zeta = 0.1:0.01:0.7;
2 ratio = 10*log10(1 ./ (4*zeta.^2 .* (1 - zeta.^2)));
3 plot(zeta, ratio)
4 xlabel('Zeta'), ylabel('Magnitude Frequency Response (dB)')
5 grid on
```

1b



```
Editor - /Users/diegoceja/Documents/MATLAB/Lab2_1d.m
Lab2_1b.m x Lab1_1c.m x Lab2_1d.m x +
1 zeta = 0.5227;
2 f0 = 4891;
3 f = 200:1:1000000;
4 h = 0.7943 ./ ((1-(f/f0).^2).^2 + 4*zeta^2*(f/f0).^2);
5 hdB = 10*log10(h);
6
7 figure;
8 semilogx(f, hdB, 'b', 'LineWidth', 1.5)
9 hold on
10
11 % Spec lines to check
12 yline(-1, 'k--', '-1 dB')
13 yline(-30, 'k--', '-30 dB')
14 yline(-55, 'k--', '-55 dB')
15
16 xline(4000, 'k--', '4 kHz')
17 xline(32000, 'k--', '32 kHz')
18 xline(180000, 'k--', '180 kHz')
19
20 xlabel('Frequency (Hz)')
21 ylabel('10log|H(f)|^2 (dB)')
22 title('Final Transfer Function')
```

1d